Lab 10: Fixed Effects, Difference-in-Differences, and Event Studies

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1 Goals

Every number below should match a slide.

1. **Panel data, quickly** (Parts 1–2): in a 2×2, four routes give the same DiD; on all years, two-way fixed effects (TWFE) with clustered standard errors; random effects and the Hausman test in one step.

2. **Staggered adoption, one number** (Part 3): see TWFE fail on the toy, decompose it on the divorce data, then get a clean single estimate with Callaway–Sant'Anna, Sun–Abraham, and imputation.
3. **Event studies** (Part 4): see that each τ_k is a 2×2, see the TWFE event study fail when cohorts differ, and run the clean event studies.

```
## install.packages(c("wooldridge", "plm", "fixest", "bacondecomp", "did",
##                    "didimputation")) once, if needed
library(dplyr)
library(ggplot2)
library(fixest)         ## feols(): fixed effects, clustered SEs, event studies,
sunab()
library(plm)            ## plm(): random effects, Hausman test
library(wooldridge)     ## the wagepan data
library(bacondecomp)    ## bacon(): Goodman-Bacon decomposition; the divorce
data
library(did)            ## att_gt(), aggte(): Callaway-Sant'Anna
library(didimputation) ## did_imputation(): Borusyak-Jaravel-Spiess
```

2 Part 1: The 2×2

```
data(wagepan)
w <- wagepan
## one row per man per year: 545 men x 8 years (1980-87); nr = man id,
## lwage = log hourly wage, married = 1 if married that year
c(men = n_distinct(w$nr), years = n_distinct(w$year), rows = nrow(w))
#>   men years  rows
#>   545     8  4360
```

As in the lecture: 1980 and 1987 only, and the men who married in between or did not change (the 17 who left marriage are set aside).

```
## each man's wage and status in 1980 and 1987, side by side
wide <- inner_join(w |> filter(year == 1980) |> select(nr, y80 = lwage, D80 =
married),
                   w |> filter(year == 1987) |> select(nr, y87 = lwage, D87 =
married),
                   by = "nr") |>
  mutate(dD = D87 - D80, dy = y87 - y80)       ## change in status, wage
  growth
keep <- wide |> filter(dD %in% c(0, 1))         ## married in between, or no
change
```

```
## long form: two rows per man; treated = married in between
long22 <- w |>
  filter(year %in% c(1980, 1987), nr %in% keep$nr) |>
  mutate(treated = as.integer(nr %in% keep$nr[keep$dD == 1]),
         post    = as.integer(year == 1987))

## Route 1: difference in differences of the four cell means
cm <- long22 |> group_by(treated, post) |> summarise(m = mean(lwage), .groups =
"drop")
m <- function(g, p) cm$m[cm$treated == g & cm$post == p]
## Route 2: interaction regression; Route 3: TWFE; Route 4: first differences
round(c(DiD_by_hand = (m(1, 1) - m(1, 0)) - (m(0, 1) - m(0, 0)),
        interaction = coef(lm(lwage ~ treated * post,
        long22))[["treated:post"]],
        TWFE        = coef(feols(lwage ~ married | nr + year,
        long22))[["married"]],
        first_diff  = coef(lm(dy ~ dD, keep))[["dD"]]), 4)
#> DiD_by_hand interaction        TWFE  first_diff
#>      0.0125      0.0125      0.0125      0.0125
```

All four are 0.0125, the slide's 0.012. With two periods, TWFE **is** the difference in differences: the man dummies remove each man's level, the year dummy removes the common change.

3 Part 2: All Men, All Years

3.1 TWFE, clustered by man

```
## all 545 men, all eight years; cluster = ~nr clusters the SEs by man
pool   <- feols(lwage ~ married,             w, cluster = ~nr)
fe_man <- feols(lwage ~ married | nr,        w, cluster = ~nr)  ## man FE only
fe_yr  <- feols(lwage ~ married | year,      w, cluster = ~nr)  ## year FE
only
twfe   <- feols(lwage ~ married | nr + year, w, cluster = ~nr)  ## both: TWFE
etable(pool, fe_man, fe_yr, twfe, keep = "married", fitstat = "n",
       headers = c("pooled", "man FE", "year FE", "TWFE"))
#>                               pool            fe_man             fe_yr
#>                             pooled            man FE           year FE
#> Dependent Var.:              lwage             lwage             lwage
#>
#> married         0.2203*** (0.0265) 0.2427*** (0.0221) 0.1501*** (0.0283)
#> Fixed-Effects:  ------------------ ------------------ ------------------
#> nr                             No                Yes                No
```

```
#> year                            No                 No                 Yes
#> --------------- ------------------ ------------------ ------------------
#> S.E.: Clustered             by: nr             by: nr             by: nr
#> Observations                 4,360              4,360              4,360
#>
#>                              twfe
#>                              TWFE
#> Dependent Var.:             lwage
#>
#> married         0.0606** (0.0215)
#> Fixed-Effects:  -----------------
#> nr                            Yes
#> year                          Yes
#> --------------- -----------------
#> S.E.: Clustered            by: nr
#> Observations                4,360
#> ---
#> Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The slides' four numbers: 0.220, 0.243, 0.150, 0.061. Always cluster by the unit: a man's wage shocks are correlated across his eight years.

3.2 Random effects, and the Hausman test

```
## plm needs the panel structure declared: id first, then time
pw <- pdata.frame(as.data.frame(wagepan), index = c("nr", "year"))
f  <- lwage ~ married + union + expersq + factor(year)
fe_plm <- plm(f, pw, model = "within")   ## fixed effects
re_plm <- plm(f, pw, model = "random")   ## random effects
round(c(FE = coef(fe_plm)[["married"]], RE = coef(re_plm)[["married"]]), 3)
#>    FE    RE
#> 0.047 0.078
phtest(fe_plm, re_plm)
#>
#>  Hausman Test
#>
#> data:  f
#> chisq = 37, df = 10, p-value = 0.00006
#> alternative hypothesis: one model is inconsistent
```

$\chi^2 = 37.0$: FE and RE differ by more than noise, so the random-effects assumption (man effects unrelated to marriage) is rejected. Report fixed effects.

4 Part 3: Staggered Adoption, One Number

4.1 The toy: TWFE gets 1.10, the truth is 1.75

The lecture's toy: three units, four periods, untreated outcome 0 everywhere. Early is treated from $t = 2$ (effects 1, 2, 3); late from $t = 4$ (effect 1).

```
toy <- data.frame(unit = rep(c("never", "early", "late"), each = 4), t =
rep(1:4, 3),
                  Y = c(0, 0, 0, 0,  0, 1, 2, 3,  0, 0, 0, 1),
                  D = c(0, 0, 0, 0,  0, 1, 1, 1,  0, 0, 0, 1))
y <- function(u, tt) toy$Y[toy$unit == u & toy$t == tt]

## the forbidden 2x2: late (treated) vs early (control), t = 2 -> 4
c(forbidden = (y("late", 4) - y("late", 2)) - (y("early", 4) - y("early", 2)))
#> forbidden
#>        -1

## TWFE, and the true average effect on the four treated cells
c(TWFE = coef(feols(Y ~ D | unit + t, toy))[["D"]], truth = mean(toy$Y[toy$D ==
1]))
#>  TWFE truth
#>  1.10  1.75
```

```
## Goodman-Bacon: TWFE = weighted average of every 2x2
## (bacon wants numeric ids, and a treatment not named D)
bt <- bacon(Y ~ post, data = data.frame(id = as.integer(factor(toy$unit)), t =
toy$t,
                                        Y = toy$Y, post = toy$D),
            id_var = "id", time_var = "t", quietly = TRUE)
bt
#>   treated untreated estimate weight                    type
#> 2       4         2     -0.5    0.2 Later vs Earlier Treated
#> 4       2         4      1.5    0.2 Earlier vs Later Treated
#> 7       2     99999      2.0    0.3     Treated vs Untreated
#> 8       4     99999      1.0    0.3     Treated vs Untreated
sum(bt$estimate * bt$weight)
#> [1] 1.1
```

The `Later vs Earlier Treated` row is the forbidden comparison: -0.5 (GB's window means) with weight 0.2, pulling TWFE down to 1.10.

```
## Callaway-Sant'Anna by hand: each block = cohort's change since g - 1 minus
the
## change of units not yet treated at t
```

```
blk <- function(u, tt, base, ctrl)
  (y(u, tt) - y(u, base)) - mean(sapply(ctrl, function(cc) y(cc, tt) - y(cc,
base)))
att <- c(`ATT(2,2)` = blk("early", 2, 1, c("late", "never")),
         `ATT(2,3)` = blk("early", 3, 1, c("late", "never")),
         `ATT(2,4)` = blk("early", 4, 1, "never"),       ## late is treated at
         t = 4
         `ATT(4,4)` = blk("late",  4, 3, "never"))
c(att, average = mean(att))
#> ATT(2,2) ATT(2,3) ATT(2,4) ATT(4,4)  average
#>     1.00     2.00     3.00     1.00     1.75
```

Every block recovers its true effect; their average is the truth, 1.75.

4.2 The divorce data

Unilateral divorce and female suicide (Stevenson and Wolfers 2006): 50 states + DC, 1964–96. `divyear` is the adoption year; 1950 means adopted before 1964 (always treated), 2000 means never adopted in the sample.

```
data(divorce, package = "bacondecomp")
dv <- divorce |> filter(sex == 2) |>            ## women
  transmute(st, year, divyear,
            y  = suiciderate_jag * 1e6,           ## suicides per million
            women
            D  = as.integer(year >= divyear),     ## 1 from the adoption year
            on
            id = as.integer(factor(st)))
table(adoption_year = dv$divyear[dv$year == 1964])
#> adoption_year
#> 1950 1969 1970 1971 1972 1973 1974 1975 1976 1977 1980 1984 1985 2000
#>    9    2    2    7    3   11    3    2    1    3    1    1    1    5

## TWFE on all 51 states, clustered by state
tw_all <- feols(y ~ D | st + year, dv, cluster = ~st)
round(coeftable(tw_all), 2)
#>   Estimate Std. Error t value Pr(>|t|)
#> D    -3.05       2.32   -1.31     0.19
#> attr(,"vcov_type")
#> [1] "Clustered (st)"
```

TWFE: −3.05, the slide's number.

```
## where the -3.05 comes from: the 2x2s by type
bd <- bacon(y ~ post, data = data.frame(st = dv$st, year = dv$year, y = dv$y,
post = dv$D),
            id_var = "st", time_var = "year", quietly = TRUE)
bd |> group_by(type) |>
  summarise(avg_2x2 = round(sum(estimate * weight) / sum(weight), 2),
            total_weight = round(sum(weight), 3))
#> # A tibble: 4 x 3
#>   type                      avg_2x2 total_weight
#>   <chr>                       <dbl>        <dbl>
#> 1 Earlier vs Later Treated    1.25         0.106
#> 2 Later vs Always Treated    -6.92         0.411
#> 3 Later vs Earlier Treated    3.36         0.254
#> 4 Treated vs Untreated       -5.21         0.228
```

The `Later vs Always Treated` and `Later vs Earlier Treated` rows use already-treated states as controls: 67% of the weight. The clean `Treated vs Untreated` comparisons average −5.2; TWFE reports −3.05.

4.3 Clean blocks, then one average

Drop the 9 always-treated states (no pre-period). The packages want a cohort variable: the adoption year, with a code for never treated (`0` for `did` and `didimputation`, a year beyond the sample for `sunab`).

```
ds <- dv |> filter(divyear >= 1964) |>
  mutate(g0  = ifelse(divyear == 2000, 0, divyear),       ## did,
  didimputation
         coh = ifelse(divyear == 2000, 10000, divyear))   ## fixest::sunab
tw_42 <- feols(y ~ D | st + year, ds, cluster = ~st)      ## TWFE, same 42
states
```

```
## Callaway-Sant'Anna: every ATT(g,t) from its own clean 2x2, then aggregate
cs_never <- att_gt(yname = "y", tname = "year", idname = "id", gname = "g0",
data = ds,
                   control_group = "nevertreated", bstrap = FALSE, cband =
                   FALSE)
cs_notyet <- att_gt(yname = "y", tname = "year", idname = "id", gname = "g0",
data = ds,
                    control_group = "notyettreated", bstrap = FALSE, cband =
                    FALSE)
## "simple": average over all treated (g, t) cells, one number
a_never  <- aggte(cs_never,  type = "simple", bstrap = FALSE)
```

```r
a_notyet <- aggte(cs_notyet, type = "simple", bstrap = FALSE)
round(c(never = a_never$overall.att, not_yet = a_notyet$overall.att), 2)
#>   never not_yet
#>  -10.56   -8.69
```

```r
## Sun-Abraham: the same blocks as one regression (cohort x event-time
dummies)
sa <- feols(y ~ sunab(coh, year) | st + year, ds, cluster = ~st)
## imputation: fit unit + year effects on untreated cells, impute Y(0),
average
imp <- did_imputation(ds, yname = "y", gname = "g0", tname = "year", idname =
"id")
```

```r
data.frame(
  estimator = c("TWFE, all 51 states", "TWFE, 42 states",
                "Callaway-Sant'Anna, never treated", "Callaway-Sant'Anna, not
                yet treated",
                "Sun-Abraham", "imputation"),
  ATT = c(coef(tw_all), coef(tw_42), a_never$overall.att, a_notyet$overall.att,
          coef(summary(sa, agg = "ATT"))[["ATT"]], imp$estimate),
  se  = c(se(tw_all), se(tw_42), a_never$overall.se, a_notyet$overall.se,
          se(summary(sa, agg = "ATT"))[["ATT"]], imp$std.error)) |>
  knitr::kable(digits = 2)
```

estimator	ATT	se
TWFE, all 51 states	-3.05	2.32
TWFE, 42 states	-0.34	2.76
Callaway-Sant'Anna, never treated	-10.56	3.12
Callaway-Sant'Anna, not yet treated	-8.69	3.40
Sun-Abraham	-10.56	3.52
imputation	-4.85	2.97

The slide's "The Estimators on One Dataset" table, plus Sun–Abraham. The clean estimators agree on the sign and say TWFE understates the effect. Callaway–Sant'Anna and Sun–Abraham with never-treated controls give the same -10.56; imputation uses every untreated year as the baseline, so it differs.

4.4 Covariates inside each block

Southern and non-Southern states may trend differently. `xformla` adjusts each block by regression (`"reg"`), weighting (`"ipw"`), or both (`"dr"`):

```r
## census region from the state abbreviation (DC counts as South)
ab <- c(state.abb, "DC"); rg <- c(as.character(state.region), "South")
ds$south <- as.integer(rg[match(ds$st, ab)] == "South")

cs_x <- sapply(c("reg", "ipw", "dr"), function(m) {
  cs <- att_gt(yname = "y", tname = "year", idname = "id", gname = "g0", data
= ds,
               xformla = ~ south, est_method = m, control_group =
               "notyettreated",
               bstrap = FALSE, cband = FALSE)
  aggte(cs, type = "simple", bstrap = FALSE)$overall.att
})
round(c(no_covariates = a_notyet$overall.att, cs_x), 2)
#> no_covariates           reg           ipw            dr
#>         -8.69         -3.68         -3.68         -3.68
```

Region is one 0/1 variable, so both models are saturated and all three agree (as on the slide's hypothetical block). Adjusting for region moves the estimate from -8.7 to -3.7.

5 Part 4: Event Studies

5.1 Each τ_k is a 2×2

One adoption date: 2 treated and 2 control units, periods 1–6, treated from $t = 4$; the effect grows 1, 2, 3 (the lecture's event-study toy).

```r
es <- expand.grid(id = 1:4, t = 1:6)
es$treated <- as.integer(es$id <= 2)
es$k <- es$t - 4                                        ## periods since the
start
es$Y <- c(1, 2, 0, 0.5)[es$id] + 0.5 * (es$t - 1) +     ## untreated outcome
        es$treated * (es$t >= 4) * c(0, 0, 0, 1, 2, 3)[es$t]
## the event study: one dummy per k for treated units, k = -1 (t = 3) omitted
round(coef(feols(Y ~ i(k, treated, ref = -1) | id + t, es)), 2)
#> k::-3:treated k::-2:treated  k::0:treated  k::1:treated  k::2:treated
#>             0             0             1             2             3

## two of them by hand, each against t = 3
mm <- function(g, tt) mean(es$Y[es$treated == g & es$t == tt])
c(tau_minus2 = (mm(1, 2) - mm(1, 3)) - (mm(0, 2) - mm(0, 3)),
  tau_0      = (mm(1, 4) - mm(1, 3)) - (mm(0, 4) - mm(0, 3)),   ## t = 4:
  already treated
  tau_1      = (mm(1, 5) - mm(1, 3)) - (mm(0, 5) - mm(0, 3)))
```

```
#> tau_minus2      tau_0      tau_1
#>          0          1          2
```

5.2 The event study on the staggered toy

Back to the staggered toy (early 1, 2, 3; late 1). The static TWFE gave 1.10. The event study gives one coefficient per k:

```
G <- c(never = NA, early = 2, late = 4)[toy$unit]
toy$k <- ifelse(is.na(G), -1000, toy$t - G)             ## -1000 codes never
treated
round(coef(feols(Y ~ i(k, ref = c(-1, -1000)) | unit + t, toy)), 2)
#> k::-3 k::-2  k::0  k::1  k::2
#>     0     0     1     2     3

## the forbidden 2x2, late vs early, t = 3 -> 4
c(forbidden = (y("late", 4) - y("late", 3)) - (y("early", 4) - y("early", 3)))
#> forbidden
#>         0
```

Exactly 0, 0, 1, 2, 3. The forbidden 2×2 is 0. Static TWFE reads it as $\beta = 0$; the event study reads it as $\tau_0 - (\tau_2 - \tau_1)$, subtracts early's growth $(3 - 2 = 1)$, and gets $\tau_0 = 1$.

5.3 What the event study still assumes

One path τ_k for every cohort. Break it: late's effect at $t = 4$ is 3, not 1. The truth by k is 0, 0, 2, 2, 3.

```
toy2 <- toy
toy2$Y[toy2$unit == "late" & toy2$t == 4] <- 3
m_toy2 <- feols(Y ~ i(k, ref = c(-1, -1000)) | unit + t, toy2)
round(coef(m_toy2), 2)
#> k::-3 k::-2  k::0  k::1  k::2
#> -0.75 -0.25  2.00  2.75  3.25
round(fixef(m_toy2)$unit, 2)                            ## late's level: 0.75,
not 0
#> early  late never
#> -0.25  0.75  0.25
```

$-0.75, -0.25, 2.00, 2.75, 3.25$: a fake pre-trend and wrong post effects, from treatment effects alone. Late jumps 3 but the shared $\hat{\tau}_0$ is 2, so OLS raises late's level to 0.75 and the leads go negative to pull its before-years back to 0.

5.4 Five event studies on the divorce data

Event time k = year − adoption year, binned at −10 and 15 for the TWFE version; never-treated states as controls (CS also with not-yet-treated).

```
ds$kb <- pmin(pmax(ds$year - ds$divyear, -10), 15)
ds$kb[ds$divyear == 2000] <- -1000                        ## never treated
## 1. TWFE event study
m_tw <- feols(y ~ i(kb, ref = c(-1, -1000)) | st + year, ds, cluster = ~st)
## 2. Sun-Abraham
m_sa <- feols(y ~ sunab(coh, year, ref.p = c(-1, -10000)) | st + year, ds,
cluster = ~st)
## 3. Callaway-Sant'Anna, averaged by k ("dynamic"), one base year g - 1
cs_u <- att_gt(yname = "y", tname = "year", idname = "id", gname = "g0", data
= ds,
               control_group = "nevertreated", base_period = "universal",
               bstrap = FALSE, cband = FALSE)
cs_d <- aggte(cs_u, type = "dynamic", min_e = -10, max_e = 15, bstrap = FALSE,
cband = FALSE)
## 3b. the same with not-yet-treated controls
cs_n <- att_gt(yname = "y", tname = "year", idname = "id", gname = "g0", data
= ds,
               control_group = "notyettreated", base_period = "universal",
               bstrap = FALSE, cband = FALSE)
cs_nd <- aggte(cs_n, type = "dynamic", min_e = -10, max_e = 15, bstrap = FALSE,
cband = FALSE)
## 4. imputation, effect by k
m_im <- did_imputation(ds, yname = "y", gname = "g0", tname = "year", idname =
"id",
                       horizon = 0:15, pretrends = -10:-1, cluster_var = "id")
```

```
## one data frame: method, k, estimate, se
grab <- function(k, est, se, m) data.frame(m = m, k = as.numeric(k), est = est,
se = se)
all <- rbind(
  grab(c(sub("kb::", "", names(coef(m_tw))), -1), c(coef(m_tw), 0), c(se(m_tw),
  0), "TWFE"),
  grab(sub("year::", "", names(coef(m_sa))), coef(m_sa), se(m_sa),
  "Sun-Abraham"),
  grab(cs_d$egt, cs_d$att.egt, cs_d$se.egt, "CS, never treated"),
  grab(cs_nd$egt, cs_nd$att.egt, cs_nd$se.egt, "CS, not yet treated"),
  grab(m_im$term, m_im$estimate, m_im$std.error, "imputation")) |>
  filter(k >= -10, k <= 15)
```

```r
## average effect over k = 0..10, by method
all |> filter(k >= 0, k <= 10) |> group_by(m) |>
  summarise(avg_k0_10 = round(mean(est), 2))
#> # A tibble: 5 x 2
#>   m                   avg_k0_10
#>   <chr>                   <dbl>
#> 1 CS, never treated       -6.53
#> 2 CS, not yet treated     -3.1
#> 3 Sun-Abraham             -6.53
#> 4 TWFE                    -3.09
#> 5 imputation              -0.97
```

```r
all$m <- factor(all$m, levels = c("TWFE", "Sun-Abraham", "CS, never treated",
                                  "CS, not yet treated", "imputation"))
ghost <- all |> filter(m == "TWFE") |> select(-m)
ggplot(all, aes(k, est)) +
  geom_hline(yintercept = 0, colour = "gray55") +
  geom_vline(xintercept = -0.5, linetype = 2, colour = "gray55") +
  geom_line(data = ghost, colour = "gray70") +
  geom_ribbon(aes(ymin = est - 1.96 * se, ymax = est + 1.96 * se), alpha = .15,
  fill = "navy") +
  geom_line(colour = "navy") + geom_point(colour = "navy", size = .8) +
  facet_wrap(~ m, ncol = 3) + coord_cartesian(ylim = c(-35, 20)) +
  labs(x = "years since unilateral divorce (k)", y = "suicides per million
  women") +
  theme_minimal(base_size = 9)
```

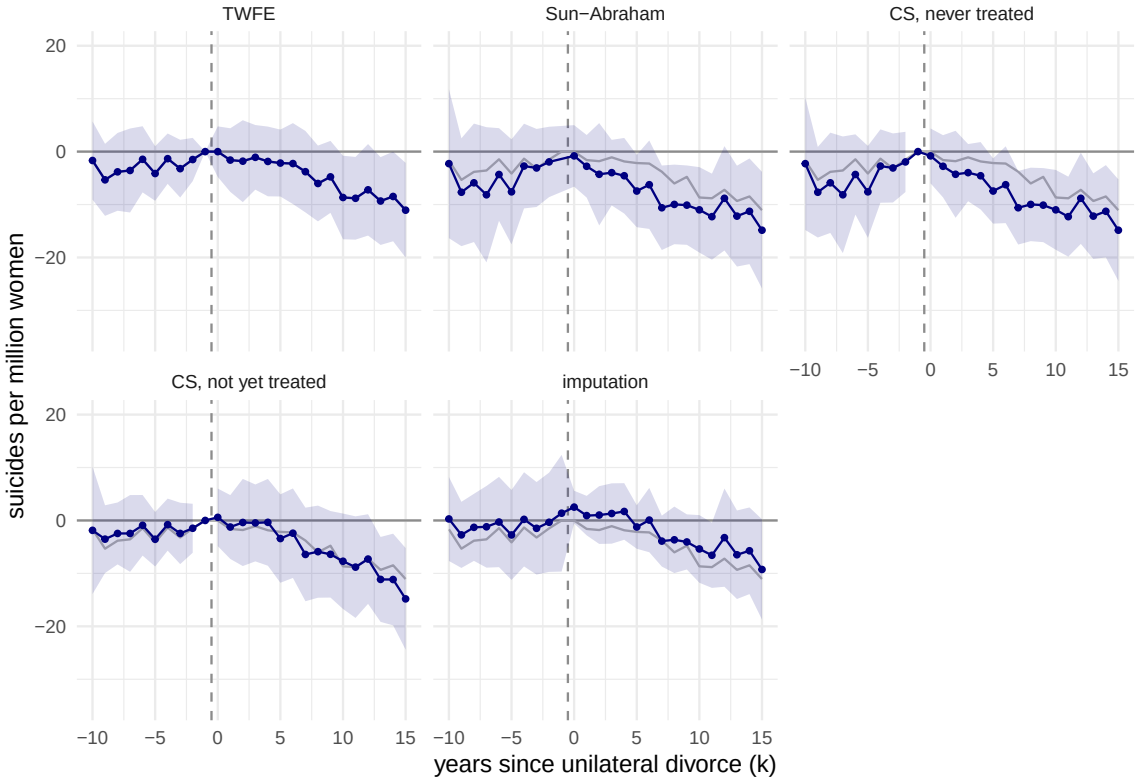


Figure 1: Five event studies on the divorce data (95% intervals). Grey: the TWFE event study, repeated in every panel.

Sun–Abraham and Callaway–Sant'Anna (never treated) give the **same** path (-6.53 on average over $k = 0$–10); the TWFE event study gives -3.09. CS with not-yet-treated controls gives -3.10: later adopters are controls early on, and by $k = 15$ only never-treated states remain (-14.82 vs. -14.85). Imputation uses every untreated year as the baseline and gives -0.97. The leads are not flat, so that choice matters.

6 Exercises

1. **The 2×2 with the 17 men who left marriage.** Put them back (`dD` in $\{-1, 0, 1\}$) and run first differences with an intercept. You should get 0.051. Why is it not $0.488 - 0.476$?

2. **Constant effects.** Change the staggered toy so every treated cell's effect is 2. Show that TWFE, the forbidden 2×2, and Callaway–Sant'Anna all give 2. In one sentence: what does the forbidden comparison need in order to go wrong?

3. **Event study vs. clean blocks.** For `toy2` (late's effect 3), compute the clean blocks for each k by hand (cohort vs. never, base $g - 1$) and check they give 0, 0, 2, 2, 3.

4. **Not yet vs. never treated.** Compare the two Callaway–Sant'Anna estimates in the scoreboard. Which uses more control states, and what extra assumption does it need?

5. **Aggregation choices.** From `cs_never`, compute `aggte(type = "group")` and `aggte(type = "calendar")`. Describe each estimand in one sentence.

7 Session information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS Tahoe 26.6.2
#>
#> Matrix products: default
#> BLAS:
/System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versio

#> LAPACK:
/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
LAPACK version 3.12.1
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: America/New_York
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] didimputation_0.5.1 data.table_1.18.2.1 did_2.5.1
#> [4] bacondecomp_0.1.1   wooldridge_1.4-7    plm_2.6-7
#> [7] fixest_0.14.2       ggplot2_4.0.2       dplyr_1.2.0
#>
#> loaded via a namespace (and not attached):
#>  [1] gtable_0.3.6        xfun_0.56           collapse_2.1.7
#>  [4] lattice_0.22-7      numDeriv_2016.8-1.1 vctrs_0.7.1
#>  [7] tools_4.5.2         Rdpack_2.6.6        generics_0.1.4
#> [10] parallel_4.5.2      sandwich_3.1-1      tibble_3.3.1
#> [13] pkgconfig_2.0.3     DRDID_1.3.0         Matrix_1.7-4
#> [16] RColorBrewer_1.1-3  S7_0.2.1            stringmagic_1.2.0
#> [19] uuid_1.2-2          lifecycle_1.0.5     compiler_4.5.2
#> [22] farver_2.1.2        maxLik_1.5-2.2      tinytex_0.58
#> [25] htmltools_0.5.9     yaml_2.3.12         Formula_1.2-5
```

```
#> [28] pillar_1.11.1       tidyr_1.3.2         MASS_7.3-65
#> [31] trust_0.1-9         nlme_3.1-168        tidyselect_1.2.1
#> [34] bdsmatrix_1.3-7     digest_0.6.39       purrr_1.2.1
#> [37] labeling_0.4.3      miscTools_0.6-30    fastmap_1.2.0
#> [40] grid_4.5.2          cli_3.6.6           lfe_3.1.1
#> [43] magrittr_2.0.4      utf8_1.2.6          bigmemory.sri_0.1.8
#> [46] bigmemory_4.6.6     withr_3.0.2         dreamerr_1.5.0
#> [49] scales_1.4.0        rmarkdown_2.30      otel_0.2.0
#> [52] zoo_1.8-15          fastglm_0.1.1       evaluate_1.0.5
#> [55] knitr_1.51          rbibutils_2.4.1     lmtest_0.9-40
#> [58] rlang_1.3.0         Rcpp_1.1.1          xtable_1.8-8
#> [61] glue_1.8.0          BMisc_1.4.9         jsonlite_2.0.0
#> [64] R6_2.6.1
```